

A Convolutional Neuro-Fuzzy Network Using Fuzzy Image Segmentation for Acute Leukemia Classification

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Abstract— Acute lymphoblastic leukemia (ALL) is a kind of blood cell malignancy that is distinguished by the presence of many lymphoid blasts in circulation. ALL accounts for over 80% of juvenile leukemia and is most common in children aged 3 to 7. The conventional approach to ALL classification requires meticulous examination of cell images by expert pathologists, which is time-consuming and results in non-standardized reports. Automation in ALL prognoses is an essential but challenging procedure. Fuzzy reasoning rules are integrated into connectionist networks when fuzzy systems and neural networks are combined to generate neuro-fuzzy systems. Current neuro-fuzzy systems, on the other hand, are created with shallow structures that have poor generalization capacity. This article automatically detects ALL from microscopic cell images, employing a deep convolutional neuro-fuzzy network. In order to reduce overfitting, data augmentations were utilized, and the number of samples got increased before training the model. Image preprocessing includes a two-stage fuzzy color segmentation technique is used to separate leukocytes from the rest of the blood components. After applying image preprocessing, the size of images decreased to 64×64 such that containing a single nucleus per image. We propose a convolutional neuro-fuzzy network based on Takagi-Sugeno-Kang (TSK) fuzzy model for acute lymphoblastic leukemia diagnosis. Our suggested model achieved an accuracy of 97.31% on average for acute lymphoblastic leukemia detection.

Keywords— *Acute lymphoblastic leukemia, Convolutional neuro-fuzzy networks, Image Classification, Image preprocessing, Takagi-Sugeno-Kang fuzzy model.*

I. INTRODUCTION

Human blood sample analysis provides data that may be used to diagnose and forecast many illnesses that, if recognized early, can be effectively treated. Acute lymphoblastic leukemia (ALL) is a type of blood cancer in which lymphocytes are produced uncontrollably and immaturately. ALL may be deadly if untreated since it spreads quickly through the bloodstream and to other critical organs in the body. Using data gathered from analyzing human blood samples, doctors can identify and cure a wide range of illnesses early on. Acute lymphoblastic leukemia

is cancer that mainly affects children under 5 and adults beyond 50 [1]. In 2010, approximately 19.3 million new cases were identified with malignant tumors, with leukemia accounting for 2.5 percent of all new cancer patients. Leukemia was responsible for 3.1 percent of tumor deaths in 2010, with the mortality rate ranking 10th among other malignant tumor deaths worldwide, according to the Global Cancer Statistics 2020 report. It is thus critical to identify the illness at the earliest possible stage to avoid additional harm to the body.

That a blood sample contains lymphoblasts indicates a clear sign to determine whether or not the patient has been diagnosed with the illness [2]. This article describes a technique for detecting lymphoblast cells in a specific blood sample, therefore reducing the risk of human error and, more importantly, aiding acute lymphoblastic leukemia early diagnosis. The ultimate objective of this research is to develop a completely automated acute leukemia classifier system that can be used as an efficient assistant to clinicians' decision-making.

Many automated leukemia diagnosis techniques have been proposed throughout the years. Four machine learning-based methods were targeted by the authors in [12], consisting of Gradient Boosted engine [9,28], Random Forest (RF), Classification and Regression Trees (CART), and C5.0 decision tree algorithm. The CART technique outperformed the other methods in their experiment. Mishra et al. [13,28] presented a combination of techniques, including a feature extraction method based on discrete orthonormal S-transform [15,28], a hybrid Principal Component Analysis (PCA), and a dimensionality reduction technique based on linear discriminant analysis. Finally, the author used an AdaBoost-based Random Forest (ADBRF) [10,28] classifier to classify reduced features. To diagnose ALL, Fathi et al. [8,28] developed an integrated technique that included neuro-fuzzy, PCA, and Group Method of Data Handling (GMDH). The authors of [14,28] utilized a combination of SENet and PNASNet which are CNN-based classification models. In order to visualize the most central portion of each cell and assess the stability of the CNN model, they employed the Grad-CAM method. Prellberg and Kramer

[25] used ResNet model [16] with Squeeze-and-Excitation modules [11] to classify leukemia cells.

Neuro-fuzzy systems are models that combine fuzzy systems with neural networks, which generally use learning methods of neural networks [31]. In problems like classification tasks and control systems, neuro-fuzzy networks are considered to be general approximators [32]. Adaptive neuro-fuzzy inference system (ANFIS) is the most well-known structures of the neuro-fuzzy model, a 5-layer feed-forward network in which a neural network was used to obtain fuzzy model parameters [33]. Since a fuzzy system has more than one rule set, neuro-fuzzy model structures are oversimplified. As with other learning structures, neuro-fuzzy models may benefit from deeper structures. Authors in [34,35] stacking TSK systems has allowed for the incorporation of deep structures into neuro-fuzzy frameworks. These models use random shift and feature augmentation to maintain outputs of each level in the same spatial position as their original inputs across networks; models have been used to categorize feature-based datasets. In [36], numerous studies on the merging of deep neural networks and fuzzy systems to classify images have been published. The extracted features using a deep neural network are applied to a fuzzy rule-based system or a fuzzy clustering in these models. However, to classify images, there have been no end-to-end neuro-fuzzy networks.

We concentrate on the identification of acute lymphoblastic leukemia using microscopic images in this work. As a result, we confront certain difficulties in identifying useful characteristics and categorization. Because of the small number of samples, the initial problem is the curse of dimensionality. With the proposed method, we will be able to solve these problems. At first, the dataset increased to 1024 images then we utilized selective filtering and unsharp masking for noise detection. After converting to Lab color space, the Gustafson Kessel clustering technique was used to extract the white blood cell nucleus from the microscopic images of blood. Using the bounding box technique, sub-images with a single nucleus per image were produced. The resulting image is considered to be the input image for the convolutional neuro-fuzzy network. The suggested network is a fuzzy classifier, which is a sort of neuro-fuzzy system. We present a convolution neuro-fuzzy system with a deep structure in the rule layer that can be applied to data with a high number of characteristics.

The remainder of the paper is organized in the following manner: Section II demonstrates the contents of the dataset in great detail. Section III outlines the whole set of preprocessing methods used on the input images, and section IV indicates the suggested convolutional neuro-fuzzy. The results are presented and discussed in section V. In section VI, the conclusion is given.

II. MATERIAL

A. Database Description

In this research, we utilized images acquired from the ALL-Image Database (IDB) data set, a publicly accessible data set that can be found online [4]. This data set has been split into two versions. There were 108 images in total for Acute lymphoblastic leukemia-IDB 1, including 59 images of healthy

individuals and 49 images of leukemia sufferers. The Acute lymphoblastic leukemia-IDB 2 data set includes 260 single cell images, 130 from leukemia patients and 130 from healthy people [19]. The 24-bit color depth images have a 257×257 pixel resolution. Fig. 1 demonstrates samples of ALL-IDB1 images that are malignant and healthy used in this study.

B. Data Augmentation

Overfitting may occur if there are not enough training samples utilized. Because our dataset had a restricted number of training examples, we used image modification to use the data augmentation method [3]. We utilized vertical and horizontal flip and image rotation to enhance the training data. Our network may be overfitted if we do not use data augmentation. The number of images was raised to 1024 using data augmentation, with 528 images representing leukemia and 496 images representing normal instances. This was done after the additional images were added. A lack of balance among the data sets used in the training process may skew the final findings. To address this problem, in each class the number of images was reduced to a bare minimum. Using this method, the total samples of the training set were split equally for both healthy and patient classes. Fig. 2 shows sample images created using data augmentation. This data set was separated into two parts: a training data set with 60% of the data and a testing data set with 40% of the data. The training data set was increased through data augmentation to minimize overfitting [19].

III. IMAGE PREPROCESSING

Using microscopic images as a starting point, preprocessing, segmentation, and classification play a crucial role in the classification process. Each image of our dataset is consists of platelets, white blood cells (WBCs), and red blood cells (RBCs). We want to separate the nucleus and cytoplasm from the remainder of the cell body by using color image segmentation to distinguish the WBC from the background. Because the cytoplasm in acute leukemia is so sparse, we have concentrated our efforts entirely on the nucleus and the information it contains. Fig. 3 demonstrates the general operating concept of the system.

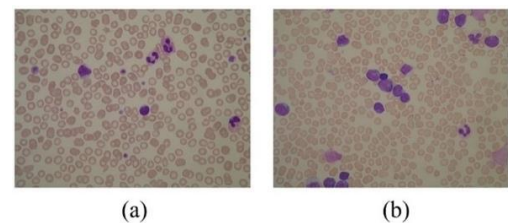


Fig. 1. A sample image (IDB-1) containing (a) normal cells (b) leukemia cells.

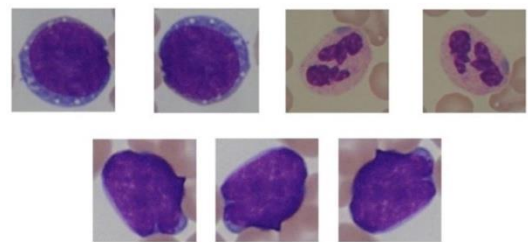


Fig. 2. Data augmentation for samples of Image DataBase (IDB-1).

A. Selective filtering and Unsharp Masking

Noise may develop during the picture collection process and due to excessive staining. Unsharp masking is applied to all images after they have been subjected to selective median filtering [5]. The ability of this method to reliably and efficiently identify the noise was improved by including an adjustable threshold in the noise detection process [27].

B. Convert RGB to Lab Color Space

Since dataset images are in the RGB color space, which makes segmenting them challenging, we utilize L*a*b* color space. Unlike RGB, which has three color dimensions, L*a*b* space has just two, so we used that to decrease the number of colors from three to two. Luminosity (L), chromaticity (a), and color (b) are considered to be three layers of the L*a*b* space. There are two parts to this representation of color information: (a) and (b). Color-based clustering is a common use of L*a*b* color space because this color space has fewer color dimensions. Before clustering, microscopic images are transformed from RGB to L*a*b* color space in the current study.

C. Image Segmentation

For extraction of the WBC nucleus from microscopic blood images, we employed color-based clustering, using a two-stage segmentation. An enhanced version of the fuzzy clustering method is used to do the first segmentation. Gustafson-Kessel clustering was utilized before L*a*b* space was used to classify objects. To generate k groups from n data, Gustafson Kessel (GK) employs a semi-supervised clustering algorithm. Fuzzy C-Means (FCM) was invented in 1973 by Dunn and enhanced in 1981 by Bezdek into a form known as GK. This technique tries to divide objects inside each cluster such that they are as close to one another as feasible while remaining as far away as possible from items in other clusters [7]. The values (a) and (b) in L*a*b* color space classify each pixel of an object into one of four clusters. The four clusters indicate cytoplasm, WBC nucleus, RBC, and stain in the background. WBC cytoplasm and RBC cytoplasm were found to be in the same cluster. The k-nearest neighbor classification algorithm is utilized in the second step of segmentation to prevent the overlaps of undesirable regions. After that, the GK clustering generates four clusters. We choose from each of those clusters a sample area at random in the second step. The mean color of each sample region is computed in L*a*b* space, and these values serve as color markers. Based on the Euclidean distance of a pixel from every color indication, any pixel in L*a*b* space is classified into one of four classes. The minimum distance between a pixel and the indicator determines the color allocated to every pixel in the image. Reconstruction of the nucleus segment is done using the labeled image [27]. We only examined the clusters containing a blue nucleus since they are needed for leukemia diagnosis.

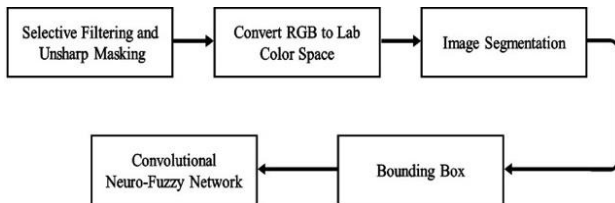


Fig. 3. Flowchart of the image preprocessing methods.

Fig. 4 demonstrates the effect of image segmentation on sample images from IDB-1.

D. Bounding Box

Each nucleus characteristic must be retrieved separately for classification as a blast cell if a reliable leukemia diagnosis is achieved. These images were generated using the bounding box method. Based on the morphology of the lymphocytes in the image, we choose just the sub-images that include them. Since the nucleus eosinophil, neutrophil, and basophil sub-images are not related with lymphocytic leukemia, they are not reviewed for classification. The application of the Bounding box method is shown in Fig. 5.

In the last part of this section, we resized images containing a single nucleus to have a resolution of 64×64 pixels.

IV. PROPOSED METHOD

This method focuses on fuzzy inference and pooling operations to implement a convolutional neuro-fuzzy network for image classification. This section discussed how these procedures were designed in the proposed network. Membership functions that map components of a universe to any values in the interval [0,1] are known as fuzzy sets. Architectures for modeling fuzzy rule-based systems are known as neuro-fuzzy systems.

A. Fuzzy Inference Operation

This method utilizes a TSK rule-based system as a fuzzy inference operation which is defined as follows:

$$\text{If } x \text{ is } A^k \text{ THEN } y \text{ is } \begin{pmatrix} f_1^k(x) \\ f_2^k(x) \\ \vdots \\ f_n^k(x) \end{pmatrix}, \quad k = 1, \dots, K \quad (1)$$

Where x , K , and n represent the input variable from the universe, the number of rules, and the number of outputs, respectively. A^k is a fuzzy set for the k^{th} rule, and f_i^k is the i^{th} output of the k^{th} rule, which are nonlinear functions of their respective inputs [26]. For image analysis, the universe and fuzzy sets are considered a set of image subregions and specific patterns, respectively. The membership matrix depicts the similarity between the subregions and given patterns.

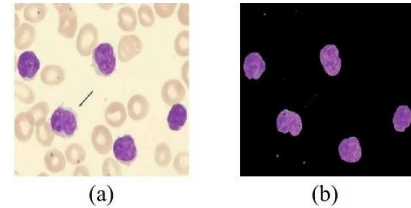


Fig. 4. Employing of two-stage image segmentation (a) Input image (b) Segmented image.

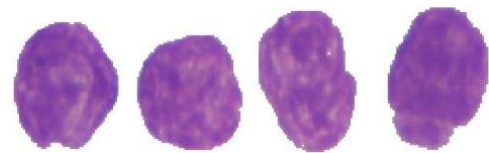


Fig. 5. Separated nucleus using the Bounding box methods.

1) *Determining the membership matrix*: The image is subdivided into subregions with the size of fuzzy sets to build the universe. In order to demonstrate the similarity between an image's subregions and patterns (A^k), the membership grades matrix (P_k) for each rule is computed. The membership values (P_{ik}) must be in the range $[0,1]$. A 4×4 one-channel input image is shown in Fig. 6, as well as 2×2 patterns for k^{th} rule. The k^{th} rule of the membership matrix is calculated for 2×2 subregions with a stride of 1. The following is demonstrated how to calculate a membership matrix:

$$p_{ik} = a_1 w_{1k} + a_2 w_{2k} + a_3 w_{3k} + a_4 w_{4k} \quad (2)$$

$$P_k = [p_{ik}] = [a_i \cdot A_k] \quad (3)$$

Where a_i and P_{ik} are the i^{th} element in the subregion and the i^{th} element in the k^{th} rule of membership matrix, respectively [26]. With the exception that the membership matrix values should be between zero and one, this approach is comparable to computing convolution operation over an image and a filter.

2) *Computing the firing strength*: For each rule, normalizing the membership matrix as follows results in firing strength:

$$\bar{P}_k = [\bar{p}_{ik}] = \left[\frac{p_{ik}}{\sum_{j=1}^K p_{ij}} \right] \quad (4)$$

Where K demonstrates the number of rules.

3) *Determining final outputs*: The following is how the final outputs are calculated:

$$y_i = \sum_{k=1}^K y_{ik} = \sum_{k=1}^K \bar{P}_k f_{ik}(x) \quad (5)$$

Where y_{ik} is i^{th} output of k^{th} filter and y_i represents i^{th} final output. The filter f is a $l \times l \times k$ filter that, after applying to $N \times N \times k$ image, returns a $N \times N \times l$ image where f_{ik} shows i^{th} element of k^{th} filter. This filter calculates the weighted sum of k resulting matrixes from the previous step for each pixel. At the end of the fuzzy inference layer (FIL), Leaky-Relu is utilized as a nonlinear activation function. The Leaky-Relu activation function can be written as illustrated below:

$$f(x) = \begin{cases} x & \text{if } x > 0 \\ 0.01x & \text{otherwise.} \end{cases} \quad (6)$$

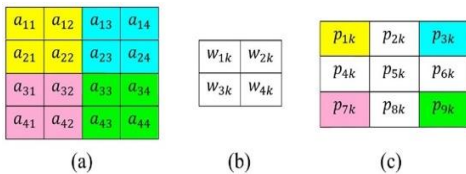


Fig. 6. Membership matrix calculation (a) The input image (b) The Fuzzy set (pattern) (c) The Membership matrix.

Fig. 7 illustrates a representation of the fuzzy inference procedure. As it can be seen, the fuzzy inference operation does not change the size of the image. TABLE I. shows the stages involved in performing a fuzzy inference. In Fig. 7, the input image is an RGB image with a dimension of $N \times N \times 3$, which, after convolution with k number of filters, k images of size $m \times m$ are obtained, where m indicates the dimension of the image after the convolution operation without padding. For each $m \times m$ acquired image, the values of the images are mapped in the range $[0,1]$ and normalized according to (4). Concatenate these images as the depth of an image with a size of $m \times m \times k$. Then we add padding to this image in height and width such that the size of the resulting image is the same as the input image. The obtained image has a size of $N \times N \times k$, which we call F_2 . Now, n number of $l \times l \times k$ filters were convolved on F_2 to produce an image of size $N \times N \times n$. The resulting image was then convolved with n number of size $n \times n$ filters. The result of these processes is known as F_3 . In contrast to this, this step involves applying n number of filters to the original picture and naming it F_4 . It is noticeable that n and m refer to the size of the patterns and the size of the image after convolving with $n \times n$ filters without padding on images with the size of $N \times N$, respectively.

B. Fuzzy Pooling Operation

Stacking the fuzzy inference operation in the layers of deep models allows layers to learn abstract attributes such as forms or specific objects. Features are recorded in their exact locations in the output of a fuzzy inference layer feature map. It indicates that simple modifications to the location of features in the input image led to new feature maps that will arise from shifting, rotation, re-cropping, and other small changes. Downsampling the feature maps is one way to deal with this issue. To address this problem the fuzzy pooling layer (FPL) is used. After the fuzzy inference layer, especially after a nonlinearity such as ReLU, a fuzzy pooling layer is applied to the feature maps created by the fuzzy inference layer. We employ average pooling in this approach, which computes the mean value for each feature patch on the feature map. The input and output of the fuzzy inference operation have the same size; however, fuzzy pooling produces smaller outputs. The filter used in the fuzzy pooling operation is lower in size than the feature map; that is usually a 2×2 pixels filter, with a two pixels stride. The fuzzy pooling layer, in other words, will always half each dimension, reducing the number of pixels to one-quarter their original size. In terms of implementation, both processes, fuzzy pooling, and fuzzy inference are comparable [26]. TABLE II. And Fig. 8 demonstrate the steps for the fuzzy pooling operation and a diagram of this operation, respectively.

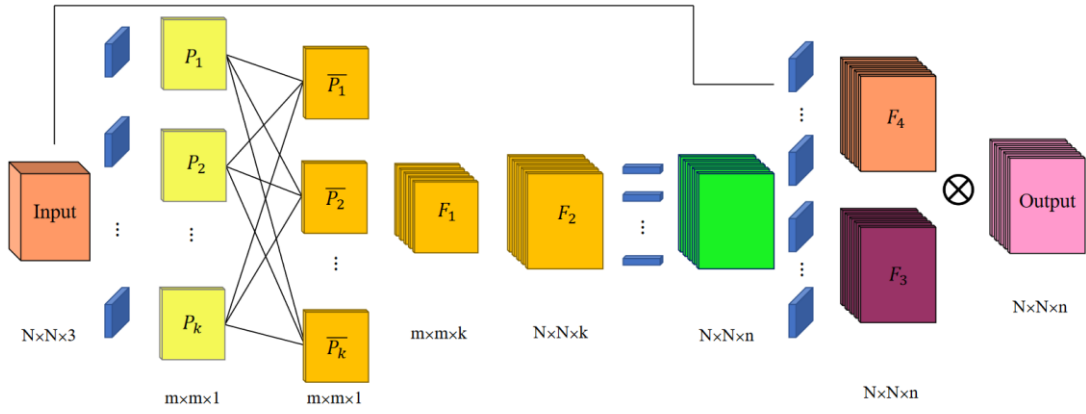


Fig. 7. Fuzzy Inference Operation based on the TSK rule-based system.

TABLE I. FUZZY INFERENCE OPERATION [26]

Steps for a fuzzy inference operation:

1. We convolve a specific filter with the input image and mapping the resulting values to the range of $[0,1]$, which calculates the membership matrix for each rule. There are as many filters as there are rules (patterns). We call this membership matrices as $P_1, P_2, \dots, P_k$.
2. We normalized the membership matrix for each rule to determine its firing strength using (4), which obtain $\bar{P}_1, \bar{P}_2, \dots, \bar{P}_k$.
3. We create $F_1 = [\bar{P}_1, \bar{P}_2, \dots, \bar{P}_k]$ by putting the resulting membership matrices together; then, with a length equal to the difference between the image size and the membership matrix, we increase the height and width of F_1 , which we call F_2 .
4. In this part, according to (5), in order to get the relationship between associated elements in $\bar{P}_1, \bar{P}_2, \dots, \bar{P}_k$, n number of $1 \times 1 \times k$ filters called f are convolved as featured maps with F_2 . There are exactly as many filters as outputs.
5. We convolved the same-sized filters as the patterns with the resulting featured maps. The number of filters is regarded as the number of outputs when applying this stage for all the outputs y_i at the same time. The result of this stage is denoted by F_3 .
6. The same size filters as the patterns used to convolve the input image. We call the obtained image F_4 . The number of filters is determined based on the number of outputs.
7. We do Hadamard product between F_4 and F_3

TABLE II. FUZZY POOLING OPERATION [26]

Steps for a fuzzy Pooling operation:

1. We convolve a specific filter with the input image and mapping the resulting values to the range of $[0,1]$, which calculates the membership matrix for each rule. There are as many filters as there are rules (patterns).
2. We normalized the membership matrix for each rule to determine its firing strength using (4).
3. We create $F_1 = [\bar{P}_1, \bar{P}_2, \dots, \bar{P}_k]$ by putting the resulting membership matrices together.
4. To obtain F_2 , we convolve 1×1 filters with F_1 .
5. The same-sized filters as the rules are convolved with the input image. The result of this stage is denoted by F_3 . The number of filters is regarded as the number of outputs.
6. We do Hadamard product between between F_2 and F_3 .

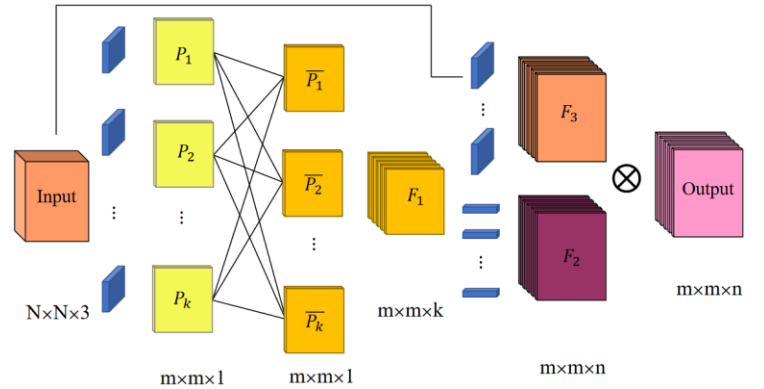


Fig. 8. Fuzzy Pooling Operation with the TSK rule-based system.

C. Convolutional Neuro-Fuzzy

The operations of fuzzy pooling and fuzzy inference are designed as chain stacks in order to create a convolutional neuro-fuzzy. Image classification involves passing feature maps to fully connected layers, which then perform the classification task on the image. To image classification, the extracted features are fed into a system of fully connected layers. Gradient descent is used to train the network. Fuzzy sets (patterns), output function ($f_{ik}(x)$) parameters, and fully connected layer parameters are all learned in the process of fuzzy pooling and fuzzy inference procedures. We use Leaky-Relu to achieve nonlinearity.

V. EXPERIMENTAL RESULTS

In order to classify the ALL-IDB1 dataset, we utilize a convolutional neuro-fuzzy network in this section. The network is consists of 2 steps. First, we chain the fuzzy inference and pooling layers and then use multilayer perceptron to classify feature maps. TABLE III. depicts the proposed convolutional neuro-fuzzy network architecture for ALL classification.

A. Chaining the Fuzzy Inference and Pooling Layers

The input of this step is 64×64 RGB images, whereas the output is $4 \times 4 \times 128$ images. To offer the images as inputs to a multilayer perceptron (MLP), max-pooling must be applied to 128 resulting images in order to obtain a 1-D feature vector with 128 features.

B. Multilayer Perceptron

The input, hidden, and output layers consist of 128, 256, and 2 neurons, respectively. We used Relu in the fully connected layers, and a Softmax layer to evaluate the likelihood of predicted classes.

We used Adam optimizer to train the network, the cross-entropy loss function and progressively lowered the learning rate across epochs. The total number of images which is 1024, was divided into training and testing data sets, with 60% and 40% of data, respectively. In order to tune the hyperparameters, we utilized 5-fold cross-validation. This method divides the observations randomly into five folds of about similar size. The first fold is considered a validation set, while the subsequent folds are considered training data. Then, using the training set, we fit a model and test it using the validation set. Finally, we keep the evaluation score of the model. Therefore the total number of images belonging to the training data set is 614. This figure is 410 for the test data set. We report the experimental results of the convolutional neuro-fuzzy network for ALL classification based on the ALL-IDB1 database. TABLE IV. shows the confusion matrix that was made for the proposed model. As it can be seen, there are very few misclassified data, and the false-negative rate is only 0.0191, which means that very few people are wrong. The false-negative rate is critical for medical diagnostics because screening a person as sick and then doing further tests to rule out the illness is preferable to incorrectly excluding a diseased person based on a false negative prediction. Furthermore, sensitivity and specificity are critical for verifying or excluding disease. TABLE V. interprets principal metrics obtained from the confusion matrix, which demonstrates the superior separability of our suggested model.

TABLE VI. illustrates the results of related works on leukemia classification.

TABLE III. FUZZY INFERENCE AND POOLING LAYERS

Type of Layer	Number of Rules	Number of Outputs	Size of Filter	Number of Stride
FIL	64	32	3×3	1
FPL	64	32	2×2	2
FIL	128	64	3×3	1
FPL	128	64	2×2	2
FIL	128	64	3×3	1
FPL	128	64	2×2	2
FIL	128	64	3×3	1
FPL	128	64	2×2	2

TABLE IV. CONFUSION MATRIX

	Actually Positive (ALL)	Actually Negative (Normal)
Predicted Positive (ALL)	205	7
Predicted Negative (Normal)	4	194

TABLE V. EVALUATION METRICS OF PROPOSED METHOD

	Accuracy	Sensitivity	Specificity	FPR
Proposed method	97.31%	0.9809	0.9652	0.0348

TABLE VI. RESULTS OF RELATED WORKS

Autor	Method	Accuracy
Mishra, 2017	Acute lymphoblastic leukemia detection based on random forest and gray level co-occurrence matrix [20,29]	96.29%
Joshi, 2013	White blood cells segmentation and classification to detect acute leukaemia [21,29]	93%
Putzu and Ruberto, 2013	Classification and detection of white blood cells [22,29]	92%
Chatap and Shibu, 2014	Support vector machine and nearest neighbor algorithm for analyzing blood samples to count leukemia cell [23,29]	93%
Putzua, 2017	Image-processing methods for leukemia detection based on leucocyte classification [24,30]	92%
Praveena and Singh, 2020	Deep convolutional neural network and sparse-FCM [18]	93.5%
Sharif, 2020	YOLOv2 Localization. PSO-based classification [17]	97.2%
Shafique and Tehsin, 2020	CNN (AlexNet) [19]	99.5%

VI. CONCLUSIONS AND FUTURE WORKS

A novel method for diagnosing leukemia has been presented in this research that evaluated ALL-IDB1 by using the convolutional neuro-fuzzy network. According to our findings, the suggested system performs better than existing deep network structures. Despite our limited data set, we achieved a high

accuracy of 97% for ALL classification by using preprocessing techniques. To avoid overfitting, data augmentations were used, and the number of samples was increased. To separate leukocytes from other blood components, we utilized an image preprocessing method comprising a fuzzy-based two-stage color segmentation algorithm. There were only two new TSK operations designed in this work. Other image classification techniques based on the TSK, Mamdani, or Zadeh models, on the other hand, can be researched in the future. This diagnostic method might also be employed in daily life if the suggested network could be trained to learn from bigger picture datasets.

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